

Marked Determinant Continuations and Singular-Safe Boundary Compression

Recovered quantum gradings, residue-four witnesses,
critical square laws, and stopping-prefix complexity

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Abstract

We develop a read-only obstruction layer for the repository's exact Bar–Natan scanning algorithm. The layer evaluates actual completed direct summands rather than only the original knot's determinant. A retained marked arc makes reduced completion functorial. Homogeneous differential entries recover the erased quantum shifts of each connected component, uniquely up to a constant, so the existing ungraded component keys suffice. The ℓ^1 norm of the quantum Euler polynomial modulo $q^4 - 1$ then gives a monotone lower bound on reduced Khovanov rank. Its matching-level values are computable by link connectivity and a signed Tait determinant, without expanding the remaining crossing cube.

For repeated determinant queries we prove a singular-safe terminal reduction. An interior block of corank r is retained as a zero block coupled to the boundary; each nonzero query uses a matrix of order at most $2b - 2$, where b is the terminal count. At the critical number $r + 1$ of terminal blocks, all nonzero quotient determinants lie in one signed rational square class. The corank threshold is sharp. Exact Python implementations, independent state-sum and homology oracles, and paired graph-kernel measurements accompany the proofs.

The resulting quasi-polynomial estimate is a theorem about successful prefixes with bounded physical representation size and frontier width. No universal bound on those parameters, unrestricted quasi-polynomial recognizer, or end-to-end production speedup is claimed.

Status. Research manuscript with written proofs and executable checks; not peer reviewed or formally verified. The package is an integration candidate, not a modification of the remote repository. Exact scope and remaining obligations are stated in Sections 1 and 8.

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1 Research target and position relative to earlier work

1.1 The bottleneck being addressed

The project in `Topology/UnknotRecognition` already has an exact scanning backend, literal direct-sum sharing, scalar-unit cancellation, and an Euler-based completed-component observer. Recent continuations have addressed twist blocks, homogeneous coefficient multiplication, radical residue profiles, and exact homological windows. Those are starting points, not results claimed again here. The reviewed component implementation stores a matching identifier, a relative homological degree, and every differential coefficient in a complete key [8, 9]. Its multiplicities are ordinary nonnegative integers, not elements of $\mathbb{F}_2$.

We address a different question:

Can one extract a stronger, inexpensive obstruction from the actual remaining diagram and the already discovered direct summands, without computing the suffix’s homology or changing the existing ungraded sharing keys?

The answer is affirmative under precise relative-complex assumptions. The observer combines the $q = 1$ and $q = i$ specializations of the reduced quantum Euler polynomial. The former is controlled by link connectivity; the latter is a signed determinant. The norm is taken *after summing within each true chain summand and before summing across independent summands*. This order prevents cancellations between independent homology groups from hiding an obstruction.

A second question is algorithmic. Many matching completions can share a large signed graph interior. Ordinary electrical-network elimination suggests a boundary response matrix, but signed weights invalidate the automatic invertibility of the interior block. We retain exactly the missing nullspace coupling. The result is a small determinant formula, a sharp vanishing gate, and a square-class law for critical terminal partitions.

1.2 What is classical and what is established here

The foundations are Khovanov’s categorification [1], Bar–Natan’s tangle formalism and local cancellation [2, 3], and the classical relation between Jones specializations and signed graph matrices; see also Boninger [5]. Schur complements and Kron reduction are established tools [6]. We do not claim these ingredients as discoveries.

The contributions developed and proved in this continuation are the following precise statements and their compatibility with this scanner:

1. Connected ungraded component keys determine quantum shift differences; therefore they also determine the completed residue norm, despite forgetting absolute quantum grading.
2. A mark-preserving completion theorem gives a monotone residue-four rank obstruction, with an exact formula for the rank it fails to see.
3. A signed terminal kernel remains valid when the interior is singular. Its corank gate is sharp, and critical partitions obey a signed square law.
4. These operations have explicit output-sensitive costs, with a quasi-polynomial consequence for controlled successful scan prefixes.

The mathematical arguments are supplied in full. A comprehensive novelty or priority search has not been completed, and no theorem is presented as an independently reviewed resolution of the unrestricted recognition problem.

1.3 Current complexity status

Lackenby’s July 2026 preprint gives a hierarchy-based incompressibility and unknot-recognition algorithm. Section 9 explicitly distinguishes estimates for the number of steps from a running-time estimate for implementations whose operations are not yet specified at that level [7, Section 9]. Our algebraic observer is not an implementation of that hierarchy algorithm.

Kronheimer–Mrowka prove that reduced Khovanov rank one detects the unknot [4]. For the early rejection result here, only the simpler fact $\dim_{\mathbb{F}_2} \widetilde{\text{Kh}}(U) = 1$ and invariance are needed. A lower bound greater than one rules out the unknot. Equality of a *lower bound* with one does not establish equality of the actual rank.

Three different meanings of “small” must remain separate. A small frontier limits the size of one coefficient algebra. A small number of matching types does not bound their multiplicities. A small observed component key does not bound the number of physically materialized intermediate objects. Our cost statement uses the latter quantity explicitly, including temporary expansion before cancellation.

1.4 Source audit and reproducibility boundary

The observed repository head was

8e5edcedb580523bdd4946233f7d0b42a97901d4.

The current component key and the Conway input were inspected at that revision. The scanner allocation and recovered-grading diagnostic were inspected at

dfc762327721d9d8040eb62a36a4ab20c005b9e5.

Per-file blob hashes and the exact scope of each read are recorded in `provenance.json`. This is a source audit, not a claim that an entire checkout of either revision was tested. The local independent package was executed; the supplied upstream diagnostic was not.

2 Recovering quantum information from ungraded components

2.1 Objects, coefficients, and weights

Fix a valid frontier with $2k$ endpoints. Matching objects have k arcs. For two objects a, b , let $c(a, b)$ be the number of circles in their overlay. A coefficient is a binary linear combination of dotted basis morphisms, indexed by subsets of these circles. If a subset S contains d circles, its intrinsic weight is

$$\omega(x_S : a \longrightarrow b) = k - c(a, b) + 2d. \quad (1)$$

The negative of this weight is the intrinsic quantum degree of the cobordism in the scanner’s convention. A nonzero homogeneous coefficient has one common dot count d across all its monomials.

Write h_v for the homological degree and q_v for the quantum shift of object v . A differential entry f_{uv} satisfies $h_v = h_u + 1$ and

$$q_v - q_u = k - c(a_u, a_v) + 2d(f_{uv}). \quad (2)$$

The repository’s grading diagnostic uses precisely this identity, rearranged as $2d = c - k + q_v - q_u$ [10].

Lemma 2.1 (Preservation of homogeneity). *Start from a genuine graded crossing complex and apply graded delooping and scalar-unit cancellations, while erasing shifts only from storage. Every remaining nonzero differential coefficient is homogeneous. A unit entry between equal matching objects is the undotted identity.*

Proof. The crossing complex and the Frobenius delooping maps are homogeneous. Composition adds intrinsic degrees. A coefficient between two fixed shifted objects can therefore contain only monomials of their prescribed degree. For equal matchings $c(a, a) = k$. If a homogeneous unit has a scalar term, that term has dot count zero and forces $q_v - q_u = 0$ in (2). Every other monomial would have positive dot count and thus a different degree. No such term can occur, so the unit is exactly the identity. Cancelling this entry is a homogeneous change of complex: the usual Schur correction is a composition of homogeneous maps of the required degree. Induction proves the assertion throughout genuine cancellations. $\square$

This lemma does not license applying the procedure to arbitrary fabricated bit matrices. The validity of the relative complex remains a precondition. The implementation checks coefficient homogeneity and potential consistency but does not reprove the provenance of the entire input complex.

2.2 The quantum-potential theorem

Theorem 2.2 (Recovery and uniqueness). *Let T be a connected nonzero-differential component of a genuine partial complex. Its matching objects and homogeneous coefficients determine all quantum shifts up to one additive integer constant. One can recover them by assigning zero to one vertex and propagating (2) along the undirected differential graph. Every remaining edge gives a consistency check.*

If two connected components have equal complete ungraded serialized keys, the key's vertex correspondence is a graded chain isomorphism after a single uniform quantum shift, in addition to the uniform homological shift already allowed by the key.

Proof. Orient each nonzero differential edge from its source to its target and give it the integer weight on the right-hand side of (2). Give the reverse edge the negative weight. Genuine quantum shifts solve these equations, so the sum of weights around every closed walk is zero. The sum along a path from the chosen root is consequently independent of the path. It is the desired potential after subtracting the root's true shift.

If q and q' are two solutions, their difference has zero change along every edge. Connectedness makes that difference constant. A one-vertex component is included: its single shift is unconstrained but immaterial up to translation.

Equal ungraded keys preserve matching objects, nonzero edges, coefficients, and relative homological degrees. In particular they preserve every weight in (2). Pulling the second component's potential back to the first therefore gives another solution, which differs by one constant. With that shift the explicit vertex correspondence preserves both gradings and every differential entry. $\square$

Corollary 2.3 (No new absolute grading key is required). *Any quantity on completed components invariant under uniform homological and quantum shifts can be evaluated from the existing complete ungraded keys. It is unnecessary to expand a shared multiplicity into its unknown absolute quantum-shift distribution.*

Proof. Theorem 2.2 identifies all copies with a chosen recovered graded representative up to precisely the shifts under which the quantity is invariant. Each copy contributes the same value. $\square$

Cost and implementation. If there are s vertices and e stored differential entries, propagation uses $O(s + e)$ graph operations after edge weights have been obtained. Homogeneity checking scans the actual coefficient support. It is not a constant-time operation on exponentially long bit-packed values. The observer therefore retains the scanner’s coefficient-size factor in its complexity bound. The function `recover_quantum_shifts` checks cycles and rejects inhomogeneous coefficients; it does not silently choose the lowest-degree term of a mixed coefficient.

3 Marked completion and residue-four rank witnesses

3.1 Why the marked arc cannot be omitted

Let $A = \mathbb{F}_2[x]/(x^2)$, with the usual Khovanov Frobenius grading. A marked point on a link gives an A -action by adding a dot to the marked component. We use the reduced complex normalized so that a single marked circle contributes one generator in quantum degree zero. Equivalently, use the quotient by the marked x -action with its conventional overall shift, or the marked- x subcomplex with the opposite normalization shift.

At scan stage i , let C_i be the processed relative tangle complex. Its actual remaining diagram defines a completion operation F_i . Choose a point on a small arc in the unprocessed suffix, disjoint from all processed operations. Denote completion followed by marked reduction by $F_i^\bullet$. A practical choice is to keep one root crossing last and mark an incident segment close to that crossing. All proper prefixes retain it.

Theorem 3.1 (Marked completion preserves relative equivalences). *Suppose every chain map and chain homotopy in a relative equivalence of processed tangle complexes is extended by the identity near the retained marked arc. Marked completion preserves that chain-homotopy equivalence and preserves finite direct sums.*

Proof. After completion, every map commutes with the dot action at the fixed marked arc: the map is the identity on a neighborhood of that arc, and the Frobenius construction is compatible with its marked A -action. The maps and homotopies are therefore A -linear. Suppose f, g and homotopies H, K satisfy

$$gf - 1 = dH + Hd, \quad fg - 1 = dK + Kd.$$

Passing to the marked quotient applies an additive functor to each identity. The same equations still hold on the reduced complexes. No assertion that the quotient functor is exact, or that its coefficient module is flat, is required: these are preserved explicit homotopy equations. Completion and the quotient both commute with finite direct sums. Uniform grading normalization does not alter the argument. $\square$

Remark 3.2 (A forbidden shortcut). It is incorrect to take arbitrary summands of a fully delooped, unmarked closed complex and divide each summand’s Euler polynomial by $q + q^{-1}$. A single surviving unmarked generator need not be a free marked A -summand. The total complex may admit a marked reduction without the chosen unmarked splitting respecting it. Keeping the mark in the suffix is what makes the direct-sum statement above legitimate. The provided observer requires a nonempty marked suffix and declines a final unmarked closure.

3.2 An integral four-coordinate shadow

For a finite bigraded cochain complex X with quantum-degree-preserving differential, define

$$\chi_X(q) = \sum_{h,j} (-1)^h \dim_{\mathbb{F}_2} X^{h,j} q^j.$$

Reduction modulo $q^4 - 1$ gives a unique vector in $\mathbb{Z}^4$:

$$\sigma_4(X)_r = \sum_{h, j \equiv r \pmod{4}} (-1)^h \dim_{\mathbb{F}_2} X^{h,j}, \quad \nu_4(X) = \|\sigma_4(X)\|_1. \quad (3)$$

These are *integer* alternating sums even though the differential is computed over $\mathbb{F}_2$.

Lemma 3.3 (Shadow norm properties). *The vector σ_4 is invariant under graded chain-homotopy equivalence and additive under direct sums. Its norm satisfies*

$$0 \leq \nu_4(X) \leq \dim_{\mathbb{F}_2} H(X).$$

A uniform homological shift changes the sign of the vector when odd; a uniform quantum shift permutes its coordinates. Both preserve ν_4 .

Proof. For each quantum degree, the alternating dimension of a finite complex agrees with that of its homology, by cancellation of boundary dimensions in adjacent homological degrees. Summing by quantum residue proves invariance. Additivity is immediate. In each residue class, the absolute value of the alternating sum is no greater than the sum of the homology dimensions; summing proves the inequality. The shift statements follow directly from (3). $\square$

Theorem 3.4 (Marked completed-component obstruction). *Assume the original input is a validated classical knot K , and a genuine relative decomposition at stage i has the form*

$$C_i \simeq \bigoplus_a \bigoplus_{h,j} T_a[h]\{j\}^{\oplus p_{a,h,j}}, \quad p_{a,h,j} \in \mathbb{N}.$$

Set $m_a = \sum_{h,j} p_{a,h,j}$. Then

$$B_4(i) = \sum_a m_a \nu_4(F_i^\bullet T_a) \leq \dim_{\mathbb{F}_2} \widetilde{\text{Kh}}(K; \mathbb{F}_2). \quad (4)$$

In particular $B_4(i) > 1$ certifies that K is nontrivial. The norm of each completed template can be evaluated using its recovered quantum potentials; absolute quantum shifts of shared copies are unnecessary.

Proof. Theorem 3.1 transports the displayed relative decomposition to a direct sum of reduced completed complexes for K , up to the irrelevant global diagram normalization. Homology of this finite direct sum is the direct sum of its homologies. Each shifted copy has the same total homology dimension and, by Lemma 3.3, the same shadow norm as its representative. Apply the norm bound separately to every copy and sum. The normalized reduced homology of the unknot has dimension one, proving rejection when the lower bound is larger. Finally Corollary 2.3 applies because the norm is shift-invariant. $\square$

Proposition 3.5 (Monotonicity under refinement). *Suppose one scan step extends each current representative additively, performs genuine relative cancellations, and splits the result into direct summands. The resulting B_4 cannot decrease while the mark is retained. Exact sharing of equal keys does not change the value.*

Proof. For a parent T , completing before or after that scan step gives homotopy equivalent reduced complexes. If the extended parent splits into children S_j with shifts, its shadow equals the sum of the shifted child shadows. By the triangle inequality, its norm is at most the sum of their norms. Multiply by the parent's nonnegative multiplicity and sum over parents. Merging equal child keys merely adds the numbers of identical contributions. $\square$

An arbitrary change of direct-sum decomposition need not be a refinement, so the proposition must not be asserted for every unrelated decomposition chosen by another algorithm. The maintained component engine's parent-local extension and splitting is the relevant situation.

3.3 Comparison with simpler observers

For $v = (v_0, v_1, v_2, v_3)$, define

$$e(v) = \left| \sum_r v_r \right|, \quad d(v) = |v_0 - v_2| + |v_1 - v_3|.$$

The first is the absolute Euler characteristic. The second is the ℓ^1 norm of the Gaussian integer obtained by evaluating at i . Both are bounded by $\|v\|_1$. Hence B_4 dominates the corresponding *reduced marked* completed-component Euler bound and the determinant character bound. This does not identify it numerically with a particular unreduced observer in the production pipeline; normalizations must be compared at the same marked level.

If a component's Euler polynomial has only one parity of quantum exponents, its four-vector has only two possible nonzero coordinates. For their sum a and difference d , its norm is

$$\left| \frac{a+d}{2} \right| + \left| \frac{a-d}{2} \right| = \max(|a|, |d|). \quad (5)$$

In general the four-vector formulation is safer than imposing a parity promise on an arbitrary formal component.

Proposition 3.6 (Exact invisible-rank formula). *Let E_r and O_r be the total homology dimensions in quantum residue r and even and odd homological degrees, respectively. Then*

$$\dim H(X) - \nu_4(X) = 2 \sum_{r=0}^3 \min(E_r, O_r). \quad (6)$$

Consequently the shadow is exact if and only if, in every quantum residue, homology occurs in at most one homological parity.

Proof. The residue coordinate is $E_r - O_r$. Subtract $|E_r - O_r|$ from $E_r + O_r$; the difference is $2 \min(E_r, O_r)$. Sum over residues. All summands are nonnegative, giving the equality criterion. $\square$

3.4 Strictness, with a realization boundary

Example 3.7 (Independent summands can reveal hidden rank). Let I be a two-ended interval object. Consider the dotted two-term complex

$$T = (I \xrightarrow{x} I\{2\})$$

in homological degrees zero and one. Complete with a marked crossingless arc. On marked reduction the dotted map is zero, so its Euler polynomial is $1 - q^2$ and its reduced rank is two. The admissible relative complex

$$I \oplus T \oplus T[1]\{4\}$$

has global residue-four norm one but split residue-four bound five. Its split ordinary Euler bound is one because each two-term block has Euler characteristic zero. The dotted blocks are connected in the differential graph before marked reduction.

This is a strict example in the relative algebraic category. No claim is made that this particular complex is the invariant of a knot tangle. In particular it is not a construction of a nontrivial knot with trivial Jones polynomial. Its purpose is to prove that the proposed norm and the position at which it is applied can matter strictly.

Safe saturation. For a rejection threshold of two on reduced rank, nonnegative multiplicities and final nonnegative contributions may be capped at two. If the existing backend supplies multiplicities capped at three, those are still safe lower bounds. Signed coordinates must be accumulated exactly until the norm is taken. Replacing $M - (M - 1)$ by independently capped terms can turn a true value of one into zero or otherwise change a subtraction; saturation is not a homomorphism on signed integer arithmetic.

There is a further distinction for a saturated scanner. If every multiplicity is capped at a fixed integer $c \geq 2$, then

$$\min\left(2, \sum_a \min(c, m_a) \nu_4(F_i^\bullet T_a)\right) = \min(2, B_4(i)).$$

Indeed, any changed nonzero product already reaches two; unchanged products are exact. Thus the threshold-capped value remains monotone. The *uncapped sum made from capped multiplicities* need not be monotone: merging two contributions of multiplicity three can replace their sum six by a shared capped multiplicity three. It is still a valid lower bound. The observer reports which monotonicity statement applies and includes the threshold-capped field explicitly. A multiplicity cap below two is rejected.

4 A polynomial-time oracle for actual matching completions

4.1 Diagram and normalization conventions

A PD crossing is a cyclic four-tuple of edge labels. Opposite slots are the same strand. The two smoothings are

$$0 : (0, 1)(2, 3), \quad 1 : (0, 3)(1, 2).$$

A closed diagram has each edge label exactly twice. The rotation system must be spherical; arbitrary four-valent incidence data are not a classical link certificate. The reference constructor

checks this using the edge involution and face permutation. If the crossing shadow has s connected components and n crossings, the spherical condition is $f = n + 2s$, checked on the resulting orientable combinatorial maps. Empty PD data additionally specify the number of crossingless components, rather than silently treating the empty monoidal unit as an unknot.

For a diagram D and state $s \in \{0, 1\}^n$, let $c(s)$ count its resolved circles and $|s|$ its number of one-smoothings. Define the raw reduced bracket

$$J_D(q) = \sum_{s \in \{0, 1\}^n} (-q)^{|s|} (q + q^{-1})^{c(s)-1}. \quad (7)$$

For a marked diagram this is its raw reduced quantum Euler polynomial. Orientation and writhe normalization multiply it by an overall signed monomial. Such factors must be tracked consistently when summing object contributions but do not change the norm of a whole completed summand. The implementation uses (7) throughout, avoiding objectwise normalizations that would introduce incompatible phases.

4.2 The value at one and exponent parity

Lemma 4.1. *Orient each component of D arbitrarily. Let $n_-(D)$ be the number of crossings whose oriented smoothing is state one in the above convention, and let $\ell(D)$ count link components. Then*

$$J_D(1) = (-1)^{n_-(D)} 2^{\ell(D)-1}. \quad (8)$$

Every exponent of J_D has parity

$$\epsilon = c(0) - 1 \pmod{2}. \quad (9)$$

Both quantities can be computed in polynomial time, without enumerating states.

Proof. At $q = 1$, switching a crossing interchanges its zero and one smoothings. Their signed contributions in (7) are interchanged with an overall minus sign. The oriented smoothing bit at that crossing also changes. Hence $(-1)^{n_-} J_D(1)$ is invariant under crossing switches, in addition to the usual normalized bracket's Reidemeister invariance. Crossing switches can make each component descending and separate the components by an ordering of their overcrossings. The result is an unlink with the same ℓ . On an ℓ -component unlink, (7) is $(q + q^{-1})^{\ell-1}$, giving (8). This is the standard $q = 1$ specialization, with its raw sign written out.

An exponent contributed by a state is $|s| + c(s) - 1 - 2j$ for some integer j . Changing one smoothing changes $|s|$ by one and changes $c(s)$ by one in absolute value. Thus $|s| + c(s) - 1$ has constant parity, namely that of the zero state. Edge pairing and opposite-slot pairing compute link components and a choice of orientations. One smoothing computation gives $c(0)$. These traversals are polynomial and in fact nearly linear with standard union-find bookkeeping. $\square$

4.3 The determinant specialization, including its phase

Assume for the moment that the crossing shadow is nonempty and connected. Checkerboard-color its spherical complement, and let G be its black Tait graph. A crossing gives one edge, including possible loops or parallel edges. For each edge e , let $b_e \in \{0, 1\}$ be the smoothing bit that *does not* join its black regions. Put

$$v = |V(G)|, \quad B = \sum_e b_e, \quad w_e = (-1)^{b_e}.$$

Let L_G be the signed weighted Laplacian: a nonloop edge of weight w adds w to both endpoint diagonals and $-w$ to the two off-diagonals. Loops are omitted because they belong to no spanning tree. A deleted root row and column are denoted by $L_G^{(r)}$.

Theorem 4.2 (Exact determinant continuation value). *For a nonempty connected classical shadow,*

$$J_D(i) = (-i)^{B+v-1} \det L_G^{(r)}. \quad (10)$$

If the shadow is disconnected, or there is an additional crossingless component, then $J_D(i) = 0$. For a single crossingless unknot it is one.

Proof. At $q = i$, the circle factor $q + q^{-1}$ is zero. Only states with exactly one circle contribute to (7).

For a subset T of black edges, its planar ribbon neighborhood has $k(T)$ components and Euler characteristic $v - |T|$. Being planar, its number of boundary components is

$$c(T) = 2k(T) - v + |T|.$$

A spanning forest inequality gives $|T| \geq v - k(T)$. If $c(T) = 1$, the displayed formula forces $1 \geq k(T)$, so $k(T) = 1$ and $|T| = v - 1$. Thus T is a spanning tree. Conversely a tree has a disk neighborhood and one boundary component. Connected classical shadows have disk faces, so this ribbon-neighborhood correspondence applies without hidden holes in a checkerboard region.

For its associated state, an edge outside T contributes smoothing b_e , and an edge inside T contributes $1 - b_e$. Therefore

$$|s(T)| = B + \sum_{e \in T} (1 - 2b_e),$$

and since $|T| = v - 1$,

$$(-i)^{|s(T)|} = (-i)^{B+v-1} \prod_{e \in T} (-1)^{b_e}.$$

Sum over spanning trees. The weighted matrix-tree identity gives their signed weight sum as $\det L_G^{(r)}$.

For completeness, the signed identity follows by writing a grounded incidence matrix E and $W = \text{diag}(w_e)$. Then $L_G^{(r)} = EWE^T$. Cauchy–Binet expands its determinant as $\sum_{|T|=v-1} \det(E_T)^2 \prod_{e \in T} w_e$. An incidence minor is zero unless T is a spanning tree, and is ± 1 for a tree, by leaf elimination. This proof uses no positivity of the weights.

Finally, different shadow components can never be joined by smoothings. Every state then has at least two circles, and the specialization vanishes. An extra crossingless component has the same effect. A single circle has reduced bracket one by definition. $\square$

The unsigned quantity $|J_D(i)|$ is the usual determinant specialization, but the phase in (10) cannot be discarded when the result is an object's contribution to a larger formal summand. The implementation uses pairs of integers for Gaussian integers; no floating-point roots of unity or numerical determinants occur.

4.4 Reconstructing the entire residue-four vector

Corollary 4.3 (Two evaluations suffice). *Let $a = J_D(1)$, ϵ be as in (9), and $d = i^{-\epsilon} J_D(i)$. Then $d \in \mathbb{Z}$ and*

$$\sigma_4(J_D)_\epsilon = (a + d)/2, \quad \sigma_4(J_D)_{\epsilon+2} = (a - d)/2, \quad (11)$$

with the other two coordinates zero. The two displayed halves are integers. Thus the residue-four vector of a completed matching can be obtained in polynomial bit time.

Proof. By parity, only the two indicated residues occur. Let their integer coefficients be u, v . Evaluation at one gives $a = u + v$, and evaluation at i gives $J_D(i) = i^\epsilon(u - v)$. Solving yields (11) and all integrality statements. The two evaluations were computed polynomially in Lemma 4.1 and Theorem 4.2; exact integer determinants have polynomial bit complexity. $\square$

Completion of a matching. The suffix PD has boundary edge labels occurring once and internal labels occurring twice. A matching pairs every boundary label exactly once. Join paired labels by union-find, relabel the suffix crossings, and validate the resulting closed PD rotation system. This constructs the actual completion at the level needed by the state sum. The observer caches the answer for each distinct matching. Nonclassical completions or inconsistent boundaries are rejected rather than assigned an arbitrary determinant.

For a component with recovered shifts q_v and homological degrees h_v , the completed vector is

$$\sigma_4(F_i^\bullet T) = \sum_v (-1)^{h_v} q^{q_v} \sigma_4(J_{D_i(a_v)}) \pmod{q^4 - 1}. \quad (12)$$

No differentials of the completed suffix are required to evaluate this Euler expression. Their existence and grading are what justify its use as a homological lower bound.

5 Singular-safe boundary compression for signed determinants

5.1 Terminal identifications and the precise algebraic problem

Let L be a symmetric integer matrix indexed by N vertices. In the graph application it is a signed weighted Laplacian. Choose an ordered nonempty terminal set $\mathcal{B}$ of size b , and let $\mathcal{I}$ be the remaining vertices. A partition π of the terminals identifies terminals in each block. Ground the block containing the first terminal.

If π has p blocks, put $q = p - 1$. Let Q_π be the $b \times q$ matrix whose columns indicate the nongrounded blocks. Terminal rows in the grounded block are zero. The grounded quotient matrix is exactly

$$M_\pi = \begin{pmatrix} I & 0 \\ 0 & Q_\pi \end{pmatrix}^T L \begin{pmatrix} I & 0 \\ 0 & Q_\pi \end{pmatrix}, \quad (13)$$

where the identity is indexed by $\mathcal{I}$. For a Laplacian this sums entries over identified vertices, with root rows and columns removed. Loops created by identification cancel in the resulting Laplacian.

Writing $A = L_{\mathcal{II}}$, the usual formula $\det M_\pi = \det A \det(Q_\pi^T S Q_\pi)$ is legitimate only if A is invertible. Signed graph interiors can fail that hypothesis even when a quotient determinant is nonzero. A Moore–Penrose inverse inserted without an additional border does not repair the determinant identity.

5.2 A maximal principal block

Lemma 5.1 (Symmetric rank-revealing elimination). *Every symmetric rational matrix A has a nonsingular principal submatrix P of order $\text{rank } A$. After permuting coordinates,*

$$A = \begin{pmatrix} P & U \\ U^T & V \end{pmatrix}, \quad V - U^T P^{-1} U = 0.$$

Such a block can be found by exact symmetric pivots of orders one and two using $O(N^3)$ rational arithmetic operations for an $N \times N$ matrix. The empty block is allowed and has determinant one.

Proof. If a diagonal entry is nonzero, use it as a one-dimensional pivot. If all diagonal entries vanish but some off-diagonal entry a_{ij} is nonzero, the principal block on i, j is $\begin{pmatrix} 0 & a_{ij} \\ a_{ij} & 0 \end{pmatrix}$, with determinant $-a_{ij}^2 \neq 0$. Symmetric Schur elimination preserves symmetry on the remaining coordinates. Continue until the residual matrix is zero. At each pivot the rank drops by precisely the pivot order, so the selected principal block has order $\text{rank } A$ and is nonsingular. Its residual Schur complement is zero. Each step updates a quadratic number of entries; there are at most N steps. $\square$

Apply this lemma only to the interior block, while updating its couplings to terminals. Let $r = \text{corank } A$. In the order selected interior, residual interior, terminals, write

$$L = \begin{pmatrix} P & U & C_1 \\ U^T & V & C_2 \\ C_1^T & C_2^T & D \end{pmatrix}. \quad (14)$$

Define

$$R = C_2 - U^T P^{-1} C_1, \quad S = D - C_1^T P^{-1} C_1, \quad \alpha = \det P. \quad (15)$$

The $r \times b$ matrix R is the essential nullspace coupling. Dropping it is exactly the error avoided by the next theorem.

Theorem 5.2 (Singular-safe terminal kernel). *For every terminal partition π ,*

$$\det M_\pi = \alpha \det \begin{pmatrix} 0_{r \times r} & RQ_\pi \\ Q_\pi^T R^T & Q_\pi^T S Q_\pi \end{pmatrix}. \quad (16)$$

If $r > p - 1$, or if RQ_π has rank less than r , then this determinant is zero. In particular $r > b - 1$ forces every terminal-partition query to vanish. Otherwise every potentially nonzero query in (16) has order at most $2b - 2$.

Proof. Eliminate P from (14) by multiplying on the left and right by mutually transposed block-unitriangular matrices. Their determinants are one. The result is

$$\begin{pmatrix} P & 0 & 0 \\ 0 & 0 & R \\ 0 & R^T & S \end{pmatrix},$$

because the interior residual $V - U^T P^{-1} U$ vanishes. Terminal identification and grounding leave the selected interior coordinates untouched. The same elimination is therefore valid in (13), with the terminal factors replaced by Q_π . Taking determinants gives (16).

If RQ_π has rank less than r , choose a nonzero vector z in the kernel of $Q_\pi^T R^T$. Then $(z, 0)$ is a kernel vector of the bordered matrix. This includes $r > q = p - 1$. A nonzero query must therefore have $r \leq p - 1 \leq b - 1$, so its order $r + p - 1$ is at most $2b - 2$. $\square$

The theorem is algebraic: it holds for any symmetric integer input, not only positive Laplacians or planar graphs. A geometric extraction theorem is a separate matter, treated in Section 6.

5.3 The critical square law

Theorem 5.3 (Critical terminal partitions have a fixed square class). *In the setting of Theorem 5.2, suppose π has exactly $r + 1$ blocks. Then*

$$\det M_\pi = (-1)^r \alpha \det(RQ_\pi)^2. \quad (17)$$

Consequently all nonzero critical-partition determinants have the same sign and the same class in $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$. For integer L , the product of any two nonzero such determinants is a positive integer square.

Proof. Now Q_π has r columns. Write $C = RQ_\pi$ and $E = Q_\pi^T S Q_\pi$. Interchanging the two groups of r columns in $\begin{pmatrix} 0 & C \\ C^T & E \end{pmatrix}$ has sign $(-1)^{r^2} = (-1)^r$ and yields the block-triangular matrix $\begin{pmatrix} C & 0 \\ E & C^T \end{pmatrix}$. Its determinant is $\det(C)^2$, including when C is singular. Equation (16) proves (17).

Nonzero squares in $\mathbb{Q}$ are positive. Thus the common sign and square class are those of $(-1)^r \alpha$. For two partitions the product is the square of the rational number $\alpha \det(RQ_\pi) \det(RQ_\rho)$. It is also an integer. A rational number whose square is an integer is itself an integer: write it in lowest terms and compare prime factors of the denominator. This proves the final assertion. $\square$

The response matrix S does not affect critical queries at all. This gives an exact shortcut, implemented in `TerminalKernel.query`, and a useful metamorphic test: changing S while preserving the other hypotheses cannot change a critical answer. The statement concerns graph cofactors. Converting it to a claim about a family of tangle closures still requires a correct common-graph extraction and the phase bookkeeping of (10).

5.4 Sharpness and a minimal counterexample

Example 5.4 (A singular interior with nonzero determinant). Take one interior vertex u and terminals t_0, t_1 , with edge weights $+1$ on ut_0 and -1 on ut_1 . In this order the Laplacian is

$$L = \begin{pmatrix} 0 & -1 & 1 \\ -1 & 1 & 0 \\ 1 & 0 & -1 \end{pmatrix}.$$

The interior block is zero. For distinct terminals, grounding t_0 leaves $\begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}$, of determinant -1 . Identifying both terminals gives determinant zero. Thus singularity of A does not make all completions vanish. This is already a planar signed graph, not an artifact restricted to nonplanar test matrices.

Proposition 5.5 (The corank gate is sharp). *For every $b \geq 2$ there is a planar signed graph with b terminals and interior corank $b - 1$ whose distinct-terminal determinant is nonzero. Hence the threshold $r > b - 1$ cannot be improved using only r and b .*

Proof. Put $r = b - 1$. Use terminals $t_0, \dots, t_r$ and interior vertices $u_1, \dots, u_r$. Add edges $u_j t_0$ of weight $+1$ and $u_j t_j$ of weight -1 . The graph is a subdivided star and hence planar. Each

interior diagonal is zero and there are no interior off-diagonals, so $A = 0_r$ has corank r . For the distinct-terminal partition, $RQ = I_r$ up to a simultaneous sign convention, and (17) gives $(-1)^r$. Equivalently, the graph itself is its unique spanning tree and has that signed product. $\square$

This proves sharpness of the *vanishing threshold*, not a lower bound on every possible representation of the determinant function. In particular one must not infer an information-theoretic necessity for a $2b - 2$ matrix from the size of the displayed bordered formula.

5.5 Batch arithmetic cost and exact bit sizes

Corollary 5.6 (Batch cost). *For a fixed symmetric integer matrix of order N , b terminals, and M terminal-partition queries, the kernel can be prepared and queried in*

$$O(N^3 + Mb^3)$$

rational arithmetic operations. After the all-zero corank test, its numeric query core needs $O(b^2)$ rational entries. Preprocessing may use $O(N^2)$ workspace; certificate and original-input storage are separate.

Proof. Lemma 5.1 gives cubic preprocessing. If $r > b - 1$, every answer is zero and the numeric query data can be replaced by that certified flag. Otherwise R and S have $O(b^2)$ entries. Since Q is a partition indicator, aggregation into RQ and $Q^T SQ$ costs $O(b^2)$ additions. The remaining determinant has order at most $2b - 2$ and costs $O(b^3)$ rational operations. Reading each partition costs $O(b)$ and is absorbed. $\square$

To turn this into a bit statement, suppose $|L_{ij}| \leq H$. Hadamard's inequality bounds a minor of order $t \leq N$ by $(\sqrt{t}H)^t$. Its bit length is $O(N(\log(N + 2) + \log(H + 2)))$. The entries in R, S are ratios of bordered minors to $\det P$; their reduced numerators and denominators therefore have polynomially bounded bit length. Exact query elimination involves minors of matrices of order $O(b)$ with those rational entries, so its bit cost is polynomial in $N, b, \log(H + 2)$ per query. No arithmetic operation on a long integer is assumed to cost one bit operation.

For signed unit-weight Tait graphs with at most n edges, one may take $H \leq n$ and $N = O(n)$, giving polynomial bit cost in the original crossing count. The reference code uses Python integers and `Fraction`, an integer Bareiss determinant for the direct oracle, and exact symmetric Schur elimination for the kernel. It retains extra index and block data for independent certificate checking, including in all-zero cases; thus the certificate-retaining object is not advertised as a minimal $O(b^2)$ -entry serialization in every degenerate case.

6 Integration contracts and executable algorithms

6.1 The general read-only observer

The package provides an observer for the stable in-memory shape of `FastScan` and `ComponentScan`. It reads matching identifiers, homological degrees, nonzero differential rows, and the algebra's matching pairs. When available, it also reads owner identifiers and multiplicity weights. It neither changes the differential nor imports the production scanner to run its local unit tests.

```

observe(valid_relative_scan, actual_nonempty_suffix, retained_mark):
    check that the mark remains in the suffix
    find connected components of the live nonzero differential graph
    for each stored component T:
        check homogeneous coefficients and recover relative q-potentials
        for each object a in T:
            complete a with the actual suffix
            validate its closed classical PD
            cache its residue-four vector using q=1 and q=i
        add the signed, q-shifted object vectors within T
        take the integral l1 norm
        multiply by T's nonnegative multiplicity
    return the sum as a reduced-rank lower bound

```

The implementation is `detshadow/continuation.py`. It explicitly rejects an absent mark, mismatched boundaries, a nonhomogeneous coefficient, an inconsistent quantum potential, and invalid differential endpoints. These checks are necessary but not a complete proof-carrying reconstruction of C_i from the original knot. An arbitrary object with scanner-like fields is not automatically a knot certificate.

The returned record distinguishes the residue-four, Euler, and determinant character lower bounds. It always reports that it does not certify an unknot. Calling code may reject a *validated knot* when the first bound is greater than one, provided the relative provenance contract has been met. A resource limit is not a topological result.

6.2 When one terminal kernel may be reused

For a batch of completed matchings, one would like a fixed signed graph G_i with terminals, and for each matching a a partition π_a and phase η_a such that

$$J_{D_i(a)}(i) = \eta_a \det L_{G_i/\pi_a}^{(\text{root})}. \quad (18)$$

A certificate for this must justify the graph correspondence and phase, not merely exhibit a numerically convenient partition. A common-disk tangle picture often motivates terminal identifications of black regions, but an arbitrary PD frontier does not automatically supply that picture. Closed regions, disconnected shadows, nesting, and the retained marked arc must all be handled explicitly.

This package supplies a conservative *verifier* for a proposed quotient identification. It independently constructs the actual completion's Tait cofactor and checks its equality, under a supplied row-and-column permutation, with the proposed quotient cofactor. The raw phase is obtained from the actual PD, not guessed from an unsigned determinant. Combined with the separate kernel block-identity verifier, this certifies the numerical bridge. It does not automatically *find* a useful fixed graph or the terminal partitions.

That verification has $O(N^2)$ matrix-comparison overhead per query, in addition to constructing the actual Tait graph. Therefore the pure $O(N^3 + Mb^3)$ batch theorem is not advertised as the cost of an automatically extracted arbitrary-PD observer. With per-query cofactor certificates the additional $O(MN^2)$ work must be charged. A proved boundary-only extractor would be needed to remove that term. The general observer in the package uses fresh actual completions and remains valid without any common-graph hypothesis.

6.3 Certificate layers

There are three logically distinct certificates:

Relative chain provenance.

The original knot diagram, scan order, valid local operations, cancellation maps, and direct-sum ancestry justify the processed complex. The present observer takes this as an input contract of the existing scanner.

Completion geometry.

A matching covers the actual suffix boundary, the completed rotation system is classical, and a retained mark makes reduced completion legitimate. For batch reuse, an additional quotient identification and phase are needed.

Exact linear algebra.

The selected principal block is nonsingular, the residual interior Schur block is zero, and R, S, α satisfy (15). The function `verify_kernel` checks these identities using a separate block solve, not by trusting the builder’s stored pivot history.

Neither a checked determinant nor a checked $d^2 = 0$ identity substitutes for the first two layers. Conversely, once those layers are valid, the observer does not need to enumerate the suffix crossing cube.

6.4 Diagnostic runner and release boundaries

The supplied `integration/upstream_probe.py` accepts a local `fast/` checkout. It observes proper prefixes of natural and shuffled orders, then completes each scanner and compares all lower bounds with its reduced rank. It also compares small cases with the independent cube, checks $d^2 = 0$, and records imported source hashes. Object exhaustion yields an explicit `UNKNOWN` record.

This diagnostic was *not run against an upstream checkout in this session*. It is not a production performance patch. In particular, the reference determinant observer is not yet interruptible inside a long integer elimination, so it must not be inserted into a hard-deadline production path without a budget policy or process boundary. The safe initial integration is opt-in, read-only observation with regression checks, not a new default recognizer.

7 Stopping-prefix complexity and the quasi-polynomial target

7.1 Execution parameters rather than an unproved structural promise

Let n be the original crossing count, and fix a scan order. Let j be a proper prefix at which a marked residue-four witness is found. Define W to be the largest frontier size among the *visited* scan stages, and let $R \geq 1$ bound the physically allocated object count at every such stage, including expansion before reduction, cancelled slots still stored in arrays, and representatives before sharing. Coefficient support sizes and exact multiplicity bit lengths are charged separately below.

These are actual execution parameters. Replacing R by the number of matching types, or by a minimal survivor count that omits temporary expansion, would not justify the same bound. Likewise, existence of a good order does not give an algorithm to find one; order generation must be supplied or included in the cost.

Theorem 7.1 (Controlled successful-prefix bound). *Assume the existing exact scanner is run by*

explicit crossing extension and ordinary finite scalar-unit elimination, with polynomial bookkeeping in its physical object count and coefficient operations costing $2^{O(W)}$ times polynomial factors. Suppose its stagewise decompositions satisfy the marked and grading hypotheses proved above. Using the general actual-completion observer at at most n visited stages adds no dependence on the maximum frontier of an unvisited suffix, and the total work through a successful prefix is bounded by

$$T_{\text{stop}} \leq \text{poly}(n, R) 2^{O(W)} \quad (19)$$

in bit operations, with an additional separately charged order-construction cost. In particular, if $W = O(\log^2(n+2))$ and $\log R = O(\log^2(n+2))$, the stopping-prefix work is $2^{O(\log^2(n+2))} = n^{O(\log n)}$.

Proof. At each visited stage there are at most R objects and $O(R^2)$ potential differential entries. At a frontier with W endpoints, an overlay has at most $W/2$ circles, so a fully stored binary coefficient has at most $2^{W/2}$ bits. The assumed finite elimination and its bookkeeping are therefore bounded by a polynomial in n, R times $2^{O(W)}$. This is a conservative bound, not a prediction of sparse performance.

Finding differential components and recovering shifts requires traversal of these explicit entries and, for validation, their coefficient supports. The same factor covers it. A matching completion retains at most n crossings and is built from polynomial-size PD data. Its connectivity, checkerboard graph, signed integer cofactor, and residue reconstruction all cost polynomial bit time in n . There are at most R distinct matching queries per observation and at most n observations.

Each crossing has two smoothings and can create at most two closed circles in a local gluing. Delooping therefore creates at most eight children per old object. The virtual expanded object count through n crossings is at most 8^n starting from the unit. Exact multiplicities consequently have $O(n)$ bits. An object-completion Euler value has at most exponential magnitude in n , and adding at most R such values uses $O(n + \log(R+1))$ bits. Recovering shifts and applying them modulo four also uses polynomial-size integers. Thus neither signed accumulation nor multiplicity arithmetic hides another uncharged exponential bit cost.

The suffix is evaluated by a determinant of polynomial order, not by scanning its crossing cube. Hence its possible scan-frontier width does not enter this observer cost. Combining these estimates proves (19). Substitution of the two logarithmic-square hypotheses proves the final statement. $\square$

The inherited Euler observer already has a direct connectivity route. The new point is not that every old suffix-width dependence is newly removed; it is that the *stronger marked determinant/residue observer* also has a polynomial completion oracle, while its erased grading can be recovered from the existing keys.

7.2 What this theorem does not establish

Equation (19) does not force B_4 to become larger than one before R or W grows. It also does not show that a nontrivial knot must be exposed by this observer while the mark remains in a proper suffix. An unknot never has such a rejection witness. It still needs a separate complete certificate or the existing exact backend. The current source and this package provide no universal bound placing all inputs in the stated quasi-polynomial regime.

A final reduced complex with zero differential can be split into individual graded generators, at which point its sum of shadow norms is exactly its rank. This mathematical observation is not a

shortcut: obtaining that complex may already have incurred exponential work. It also differs from illegally dividing arbitrary unmarked closed summands, as warned in Remark 3.2.

For completeness, full reduced rank one over $\mathbb{F}_2$ still suffices for unknot recognition. Universal coefficients bound the reduced rational rank by the $\mathbb{F}_2$ dimension. The rational rank is at least one, for instance by its normalized Euler value at one. Thus $\mathbb{F}_2$ rank one forces rational rank one, and Kronheimer–Mrowka apply [4]. None of these implications says that a partial lower bound of one is sufficient.

7.3 A reasonable performance policy

The observer should initially be used where the maintained algorithm has already discovered several distinct summands or where cached completions make its cost modest. An optional policy can skip a probe after its budget is exhausted and continue the original scanner. Such skipping loses a possible early certificate but cannot change the represented complex. The lower bound from an earlier successful probe remains valid.

To evaluate actual improvement, measure the added determinant cost, the first stage exposing $B_4 > 1$, and the remaining scan work avoided. Reporting only a faster determinant kernel would omit the decisive end-to-end question. Conversely, large virtual multiplicities can make a small positive template contribution decisive, but only after that contribution has been justified by marked completion.

8 Validation, measurements, and reproducible artifacts

8.1 Independent checks actually executed

All computations in the delivered local package use the Python standard library. The independent cube is an exponential validation oracle and is not called by the fast matching-specialization path. The completed checks are summarized in Table 1; full inputs and outcomes are retained in the JSON records.

Check	Completed count
Unit-test methods	35
Diagram shadow comparisons	701
Explicit smoothing states in those comparisons	133,923
Reduced homology comparisons	83
Columns checked for $d^2 = 0$	62,904
Signed random graphs	160
Graphs with singular interior blocks	57
Terminal-quotient determinant comparisons	4,000

Table 1: Executed local checks. Counts refer to the delivered run, not to an upstream production test suite.

The diagram experiment uses seed 20261007 and 700 generated braid closures on two through six strands with at most ten crossings, including links, unused strands, and disconnected shadows. It adds the repository’s 11-crossing Conway PD [11]. Every fast residue vector is compared with

the full state sum, which independently expands the factors $(q + q^{-1})^{c(s)-1}$. A bounded subset is compared with a separately built marked reduced differential and exact binary homology. In that oracle the marked circle is forced to carry x , with its conventional reduced grading shift. Merge and split maps are implemented directly from the Frobenius rules; every generated two-step differential column is checked.

The Conway input has reduced $\mathbb{F}_2$ rank 33 in the independent computation, while its global residue-four norm is one. The positive three-braid $(\sigma_1\sigma_2)^5$ has reduced rank seven and again global residue norm one. These are controls demonstrating the incompleteness of the global specialization. They do *not* demonstrate an early component-stage speedup on these knots, because no upstream scan of them with the observer was executed here.

The graph experiment compares every kernel query with an independently constructed quotient matrix and an integer Bareiss determinant. It includes singular interiors and checks the kernel by an independent block-identity solve. Unit tests additionally cover zero-dimensional determinants, two-dimensional pivots with zero diagonals, all partitions of small terminal sets, the sharp-corank family through $r = 12$, the critical square law, coefficient homogeneity, potential cycle failures, and deliberately corrupted algebraic or geometry certificates.

Finite computations support the implementation and expose normalization errors; they are not substitutes for the written proofs. Passing these checks is not a Lean formalization, an exhaustive theorem-priority review, or a complete topological certificate for arbitrary externally supplied scanner objects.

8.2 Paired exact determinant-kernel benchmark

The benchmark compares two fresh direct controls, A and B, with a compressed arm. Each direct arm rebuilds every quotient matrix and computes its exact Bareiss determinant. The compressed arm includes both kernel preprocessing and all its queries. Graph construction precedes the timed arms; independent certificate verification is excluded from all arms. Arm order is randomized in each of three paired rounds. Every integer answer is checked for equality. Each case has six terminals and 96 queries.

N	r	Direct A (s)	Compressed (s)	A/compressed	A/B
32	0	0.1054	0.0189	5.52	1.002
56	0	0.5727	0.0830	6.90	1.011
80	0	1.8632	0.2928	6.43	1.031
56	1	0.6086	0.0864	7.05	0.971

Table 2: Exact determinant-kernel measurements. Times are medians; ratios are medians of paired ratios, not ratios of medians. The compressed arm includes preprocessing. Each case uses six terminals and 96 queries.

The measurements are from CPython 3.13.5 on the recorded Linux x86-64 host. They are small-sample, host-dependent kernel measurements, not fitted asymptotic exponents or hardware-independent guarantees. The A/A ratios measure ordinary timing variability; three rounds are not a confidence interval. The signed graph family is not asserted to be the frontier graph of a measured knot scan. In particular the table does not establish that recognizing a knot is faster by its listed factors.

8.3 Package map and reproduction

The package contains the self-contained article source and compiled PDF, six implementation modules, unit tests, complete validation and benchmark JSON, a worked singular-kernel certificate, source provenance, and integration notes. The important paths are:

Path	Purpose
<code>detshadow/continuation.py</code>	Quantum recovery and read-only component observer.
<code>detshadow/diagram.py</code>	PD validation, signed Tait phase, residue oracle.
<code>detshadow/linalg.py</code>	Exact determinants, singular kernel, independent verification.
<code>detshadow/geometry_certificate.py</code>	Proposed quotient-to-actual-PD verifier.
<code>detshadow/cube.py</code>	Independent reduced homology validation oracle.
<code>experiments/</code>	Seeded comparisons and paired kernel benchmark.
<code>integration/upstream_probe.py</code>	Local-checkout diagnostic, supplied but not executed here.
<code>results/</code>	Raw evidence, not merely aggregate speed claims.

From the package root:

```
python -m unittest discover -s tests -v
PYTHONPATH=. python experiments/validate.py
PYTHONPATH=. python experiments/benchmark.py
cd paper && sh build.sh
```

No knot-library dependency is required. Running the experiments overwrites the corresponding JSON with fresh local measurements; the delivered PDF records the delivered run. Preserve those records before producing a comparison. The article’s TeX file embeds its tables and bibliography, so it can also be compiled independently.

9 Further research questions with concrete success criteria

9.1 Immediate correctness and integration questions

1. Complete upstream prefix regression. Run the supplied diagnostic on a pinned complete checkout, with natural, greedy, and shuffled orders. The immediate success criterion is agreement with full reduced ranks, valid recovered potentials at every tested stage, and monotonicity whenever the component decomposition evolves by refinement. Tests should include determinant-one knots and RI/RII-expanded unknots. A failed potential equation should be diagnosed as a grading or interface failure, not bypassed.

2. A certified terminal-partition producer. Construct a map from a suitable frontier embedding to one signed suffix graph and the partition and phase for every matching. Prove exactly when black-region identifications are sufficient and when extra region or connectivity data are needed. A useful result would give $\text{poly}(b)$ per-matching extraction and a checkable embedding certificate, rather than only an $O(N^2)$ cofactor comparison. Arbitrary PD frontiers must not be silently treated as one disk.

3. Interruptible exact arithmetic and inference budgets. Add cancellation-safe interruption points or isolate determinant probes in bounded worker processes. Record preprocessing, rational bit sizes, coefficient support scans, and observer time separately from scanner time. The success criterion is that exhausted optional inference leaves the original complex untouched and never becomes an unknot verdict.

9.2 Questions about exposing useful witnesses

4. Early residue witnesses for determinant-one knots. On the repository’s Conway, Kinoshita–Terasaka, and determinant-one stress inputs, determine whether literal component splitting exposes $B_4 > 1$ substantially before the complete backend. A positive result needs paired end-to-end measurements and the associated stage records. A negative result would be equally informative if it identifies why summands remain coupled.

5. Root and order optimization. The retained root crossing constrains the final part of an order. Can one choose a root and order jointly to minimize observed cost until a witness appears? A proved target would be a constructive family theorem, not just the existence of a good order found by exhaustive search. Certificate quality and frontier width need not be optimized by the same order.

6. Coherence after local simplification. Proposition 3.6 gives an exact target: eliminate overlap between even and odd homological parity in each quantum residue of completed summands. Identify a nontrivial class of genuine knot tangles for which local simplification guarantees this coherence. Such a theorem would turn the residue observer from a lower bound into an exact rank computation on that class. The abstract dotted-block example is not such a realization theorem.

9.3 Stronger algebraic and structural questions

7. Dynamic singular kernels. As the scan moves, the suffix graph changes. Can P, R, S be updated under crossing deletion and terminal promotion without rebuilding from scratch, while certifying rank changes and preserving polynomial bit bounds? An algorithm assuming constant corank would need a separate mechanism for rank-drop events; those are precisely where an ordinary inverse update fails.

8. Topological meaning of the critical square class. When a fixed-graph tangle family satisfies the quotient contract, the critical-partition determinants share one rational square class. Determine which tangle families realize large corank and whether their square-class constraint yields useful pruning before exact determinant computation. Any statement about link determinants must remove raw phases correctly and handle zero determinants separately.

9. Higher cyclotomic residue observers. Replace $q^4 - 1$ by another explicitly tractable quotient. The algebraic norm proof extends immediately, but the matching-completion oracle may no longer be polynomial-time. The substantive target is a stronger observer with a proved exact evaluation algorithm and bit-cost bound, not simply more sample points evaluated exponentially.

10. Physical-size bounds and witness exposure. Prove, for a meaningful family of input diagrams, that a useful marked witness is exposed while W and $\log R$ stay polylogarithmic. This is the missing bridge to a family-level quasi-polynomial recognition theorem. Bounds on coefficient algebra dimension or matching-type count alone do not supply it. A strong negative result would construct families forcing large physical representation before every witness in this observer class.

11. Combining windows, twists, and component observers. The new observer can in principle share a valid intermediate complex with the existing window and twist mechanisms. A composition theorem must track which guard degrees, marked arcs, and decomposition maps survive the handoff. A claimed combined speedup should be measured against each existing mechanism separately as well as against the default pipeline.

12. Formal proof and minimal certificate development. The matrix identities, square law, residue norm properties, and graph potential uniqueness are good initial formalization targets. The harder step is connecting scanner operations to a marked graded tangle category. A small numerical certificate is not enough until that semantic bridge is also trusted or verified.

10 Conclusion

The completed continuation supplies a stronger exact observer and a singular-safe determinant engine. Its most useful interface result is that homogeneous coefficients recover enough quantum information from existing ungraded connected-component keys. Its most concrete new algebraic consequence is the critical-partition square law, together with a sharp corank gate and a batch determinant formula valid for signed weights.

These results isolate a tractable portion of the recognition problem: completed summand obstructions can be evaluated without expanding the suffix cube, and shared graph interiors can be compressed without assuming invertibility that signed graphs do not possess. The remaining global question is whether genuine knot computations expose decisive summands before physical representation growth becomes large. The package provides proofs, executable checks, and explicit integration boundaries for studying that question, rather than treating a local kernel improvement as a proof of unrestricted quasi-polynomial recognition.

A Worked phase and square-law examples

A.1 A trefoil state-sum check

For the package’s positive two-strand word $(\sigma_1)^3$, the raw reduced polynomial computed by the independent state sum is a signed monomial normalization of the usual trefoil polynomial. Rather than making an orientation-dependent table a convention, the test checks the complete Laurent dictionary against the signed Tait phase and against all four residues. The reduced rank is three, the absolute Euler value at one is one, and the determinant character norm is three. Thus the initial residue observer already detects this knot, although the production pipeline’s existing determinant filter would do so more cheaply. This example validates normalization; it is not evidence of a new pipeline advantage.

A.2 A square-class family with a nontrivial scale

Start with r balanced interior vertices: each is incident only to terminals, and the sum of its signed incident weights is zero. Thus their interior block is zero. Add an additional interior leaf attached to the root terminal by a weight-two edge, or equivalently by two parallel unit edges. Its selected principal block is $P = (2)$, so $\alpha = 2$. For any critical partition into $r + 1$ terminal blocks,

$$\det M_\pi = (-1)^r 2 \det(RQ_\pi)^2.$$

In this construction R is integral. The nonzero absolute determinants are twice integer squares, regardless of arbitrary additional terminal edges in the response block. This is the family used to test the critical shortcut for $r = 1, 2, 3$ with five terminals and all critical partitions. No factorization algorithm or probabilistic square test is involved in the formula.

B Proof and implementation failure checklist

Tempting shortcut	Required correction
Discard quantum information completely	Recover edge-weight potentials within each connected component; check homogeneity and cycles.
Use arbitrary unmarked closed summands	Retain a mark outside processed operations before taking reduced completion.
Use $p_a(-1)$ as multiplicity	Count copies nonnegatively with $p_a(1)$; shifts act only before a shift-invariant norm.
Saturate signed Euler coordinates	Accumulate exactly; cap only nonnegative contributions after the norm.
Replace a complex by disconnected vector records	Require genuine relative decomposition and ancestry, not only the appearance of a sparse matrix.
Drop the phase of $J_D(i)$	Retain $(-i)^{B+v-1}$ until all terms of a component are summed.
Use a Tait cofactor on a disconnected shadow	Return the zero specialization, or justify a separately correct disconnected formula.
Set all queries to zero when A is singular	Keep the residual nullspace coupling R ; use the bordered formula.
Use a pseudoinverse as an ordinary inverse	Prove the relevant singular determinant identity; the missing border cannot be omitted.
Apply a batch graph kernel to every matching	Establish the fixed-graph quotient and phase contract for that geometry.
Call a parameterized stopping bound universal	Prove constructive bounds on visited physical size and witness exposure for the intended input class.
Report kernel speedup as recognition speedup	Run the complete opt-in and baseline pipelines with paired measurements.

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